Countercyclical Consumption Taxes

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  - Transfers better to boost private consumption, but output multiplier $\approx$ MPC on impact

    Kaplan & Violante (2014), McKay & Wolf (2025), Ferriere & Navarro (2025), Ramey (2025), ...

Policy Proposal

- Alternative instrument to mitigate recessions: Consumption tax cuts
  - Proposed in the policy debate, used in some countries: UK 2009, Germany 2020

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- HANK model to quantify business cycle stabilization
  - Compute multipliers, investigate welfare

- Roadmap: Theoretical equivalence; Quantitative model

Literature

- Effects of monetary policy and government spending in HANK models

  Kaplan, Moll, and Violante (2018), Hagedorn, Manovskii, and Mitman (2025), Bilbiie (2020), Auclert, Rognlie, and Straub (2023), Ferriere and Navarro (2024), Alves and Violante (2023)

- Optimal fiscal and monetary policy in HANK

  Bhandari, Evans, Golosov, and Sargent (2021), Le Grand, Martin-Baillon, and Ragot (2024), McKay and Wolf (2023) Broer, Druedahl, Harmenberg, and Oberg (2025), Le Grand, Ragot and Bourany (2025)

- Stabilization and consumption taxes in HANK

  Parodi (2024), Bachmann, Born, Goldfayn-Frank, Kocharkov, Luetticke, & Weber (2025), Bartal & Becard (2025)

- Equivalence results in HANK

  Correia, Nicolini & Teles (2008), Correia, Farhi, Nicolini & Teles (2013), Seidl & Seyrich (2023), Wolf (2024), Wolf (2025)

- Evidence on consumption tax cuts

  Blundell (2009), Benzarti, Carloni, Harju, & Kosonen (2020), Bachmann et al. (2025)

A Theoretical Equivalence

Theory: Roadmap

1. Equivalence result between a consumption tax cut and a spending shock
   - **Key assumption: preferences**
   - Parsimonious model: representative agent, no capital, no rigidities
   - Extension to full HANK model

2. Departing from equivalence: changing preferences
   - NK model with a representative agent and non-distortionary taxes

A Parsimonious Model

- Representative household
  - Values consumption C_t and leisure $1 - N_t$, can save in a risk-free bond A_t
- Representative competitive firm
  - Hires labor in a competitive labor market to produce output Y_t
- Government has to finance public good G_t
  - Uses labor taxes, lumpsum taxes, and consumption taxes τ_t^n, T_t, and τ_t^c
  - Issues public debt $B_t = A_t$ in equilibrium
- Prices: (r_t, w_t) the interest rate and the net-of-taxes wage

Two Policies

- From steady state, consider two temporary fiscal policies.

1. Government spending shock $\left\{\hat{G}_t > G\right\}_{t=0}^{T}$
   - Financed with any policy $\hat{\Pi}_t \equiv \left(\hat{\tau}_t^n, \hat{T}_t, \hat{B}_t\right)$ converging back to steady state
   - Define $\hat{F}_t \equiv \hat{G}_t - \tau^c \hat{C}_t > F$ the fiscal cost
   - Let $\hat{X}_t \equiv \left(\hat{Y}_t, \hat{r}_t, \hat{w}_t\right)$ be the vector of output and prices

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2. Consumption tax cut $\left\{\bar{\tau}_t^c < \tau^c\right\}_{t=0}^{T}$
   - Financed with any policy $\bar{\Pi}_t \equiv \left(\bar{\tau}_t^n, \bar{T}_t, \bar{B}_t\right)$ converging back to steady state
   - Similarly, define $\bar{F}_t \equiv G - \bar{\tau}_t^c \bar{C}_t > F$, and $\bar{X}_t \equiv \left(\bar{Y}_t, \bar{r}_t, \bar{w}_t\right)$

Equivalence Result

- **Assumption:** Assume log-separable preferences $U(c, n) = \log c - v(n)$.

- **Proposition:** Consider any spending shock of fiscal cost $\hat{F}_t$ financed with policy $\hat{\Pi}_t$.

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  - The government's budget constraint will hold;
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- **Corollary:** The two policies generate identical output multipliers as $dY^g = dY^\tau$, but different consumption multipliers as $dC^g < dC^\tau$

Intuition **The log case**

- A recursive formulation of the household problem reads:

$$V_t(A_t) = \max_{C_t, N_t, A_{t+1}} \{\log(C_t) - v(N_t) + \beta V_{t+1}(A_{t+1})\} \text{ s.t.}$$

$$C_t = [w_t N_t + (1 + r_t)A_t - A_{t+1}] / (1 + \bar{\tau}_t^c) \quad \text{(BC)}$$

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  $\Rightarrow$ If taxes and prices are identical, expenditures are identical:

$$\bar{e}_t \equiv (1 + \bar{\tau}_t^c)\bar{C}_t = (1 + \tau^c)\hat{C}_t \equiv \hat{e}_t$$

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- Aggregate supply will also be identical across the two policies

Extending the Equivalence

- Straightforward extension to aggregate shocks
  - **Equivalence** of any package $\{\tilde{G}_t, \tilde{\tau}^c_t\}$ delivering the same fiscal cost:
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- Key assumption: **log preferences**, income and substitution effects cancel out
  - Static margin (labor supply)
  - Dynamic margin (savings decisions)

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Beyond log separable preferences

- Representative-agent NK model, fiscal policy financed with lump-sum transfers
  1. Flexible prices: r_t so that the Euler equation holds
  2. Nominal rigidities and central bank implementing constant real rate
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    where κ is the slope of the Phillips Curve (small)
  - (Consumption responses: $dC^g \leq 0 < dC^\tau$).

Quantitative Model

A HANK Model with Unemployment

- Households
  - Heterogeneous in income and wealth
  - Heterogeneous MPCs
- Unemployment risk and Okun's law
  - Unequal incidence across households and increasing in recessions
- Nominal rigidities
  - Aggregate demand effects

Model Overview

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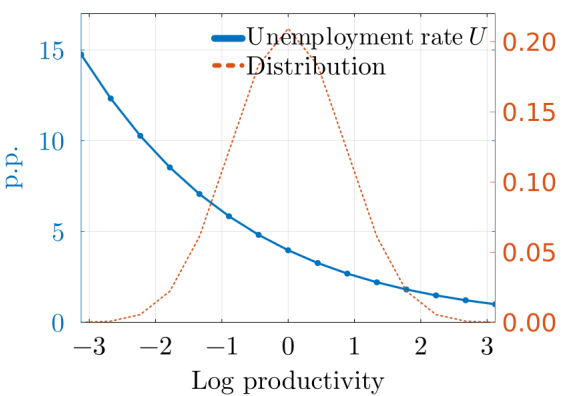
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  - Sticky wages: Labor unions choose homogenous level of labor supply
  - Monetary authority implements a standard Taylor rule

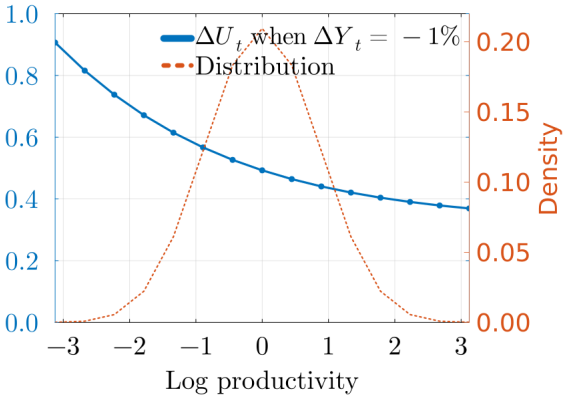
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- Government
  - Finances spending, transfers, UI benefits with debt, and labor, consumption, capital taxes

Calibration

Unemployment: Steady State and Business Cycles





Calibration **Household consumption responses**

- Marginal propensities to consume (mpc)

  Parker, Souleles, Johnson, and McClelland (2013), Kaplan and Violante (2014), ...
  Saporta-Eksten (2014), Ganong and Noel (2019)

  - Compute mpc out of a \$500 rebate: average quarterly mpc at 0.25
  - Decline with wealth: from 0.72 to 0.03 from 1st to 5th wealth quintile

Wealth quintile	1	2	3	4	5
Wealth share	0.00	0.01	0.04	0.15	0.80
mpc	0.72	0.22	0.10	0.06	0.03

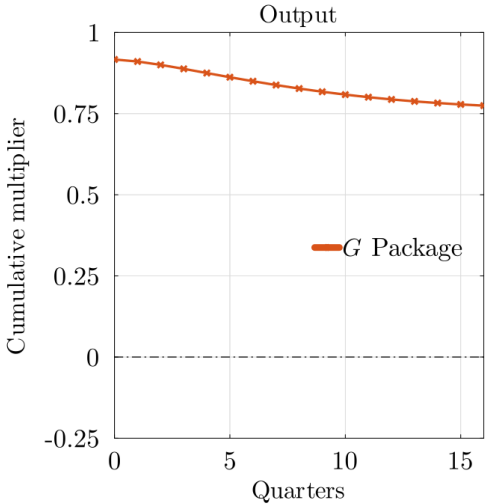
  - Larger for unemployed at 0.59, consumption drops by 6% when falling into unemployment
    Saporta-Eksten (2014), Ganong and Noel (2019)

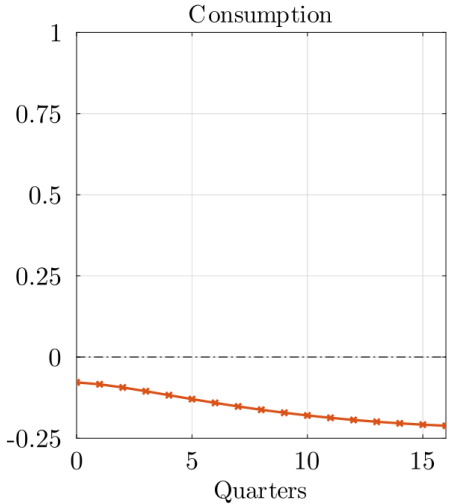
Spending, Checks, and Consumption Taxes

Three Fiscal Packages

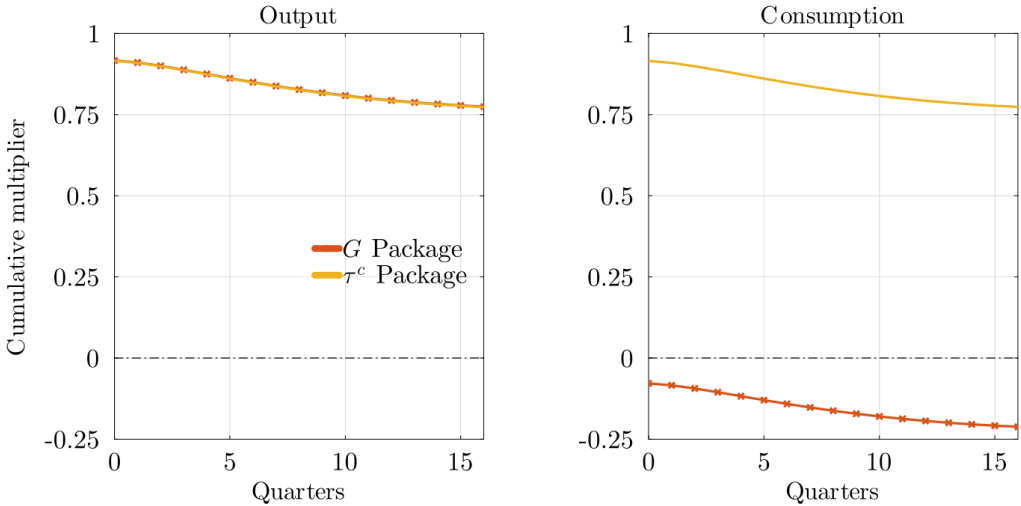
- Three packages of equivalent cost
  - Government spending shock with persistence $\rho_w = 0.75$
  - Consumption tax cut, same size & persistence
  - Lump-sum check, same size & persistence

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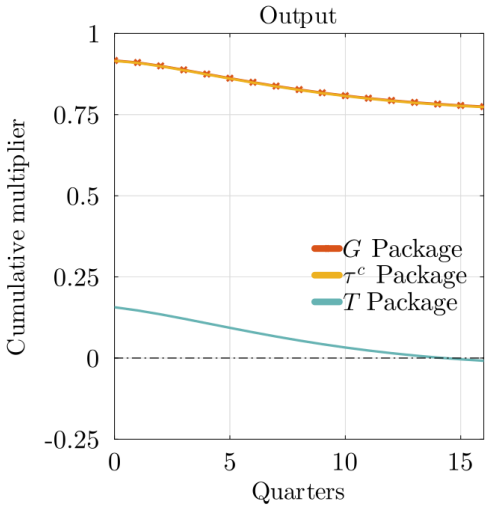


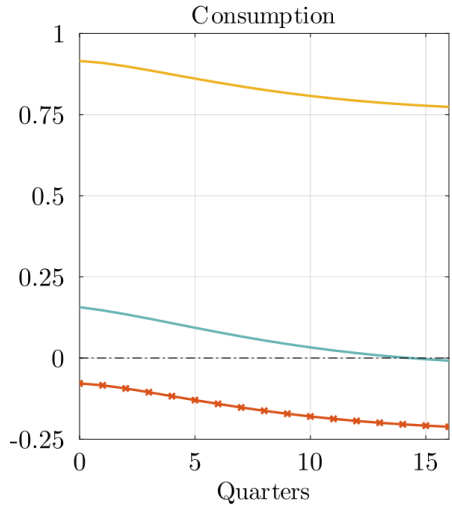


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More Fiscal Packages

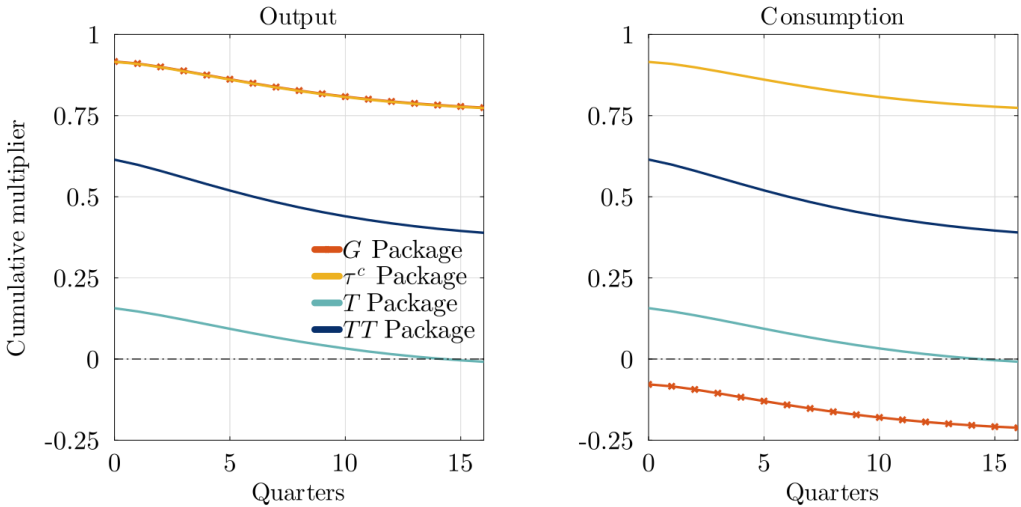
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- Targeted transfers
  - Functional form based on logistic function
    Ferriere & Navarro (2025)

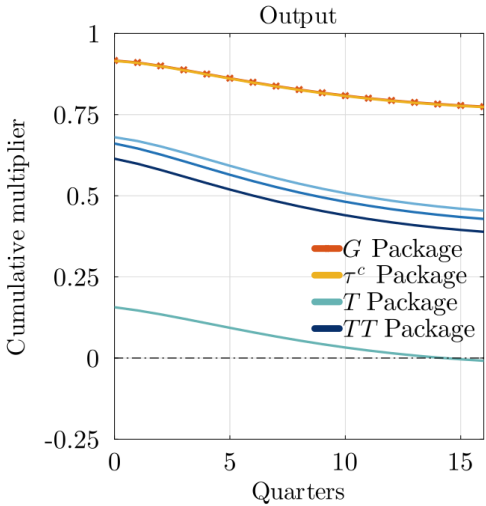
$$TT_t(y) = m_t \frac{2\exp(-\xi y/\bar{y})}{1+\exp(-\xi y/\bar{y})}$$

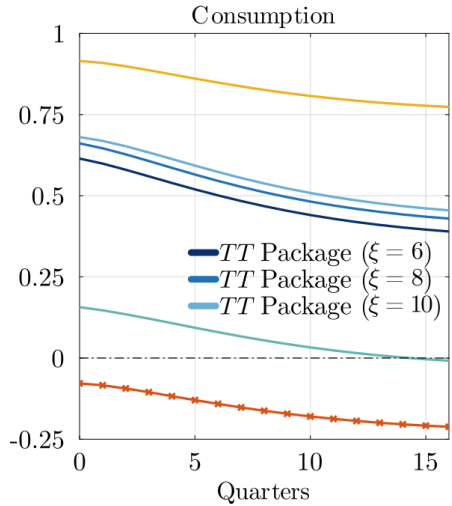
  - Baseline: Bottom-20% receives 70% of the transfers; Robustness: 80%, 90%

Three Fiscal Packages



Three Fiscal Packages





Recession and Welfare

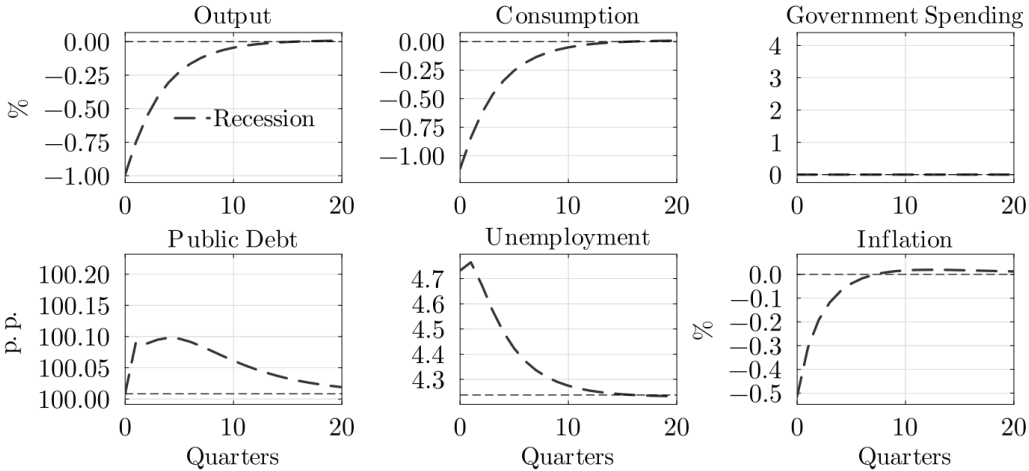
Recession **Demand Shock**

- Recession induced by a negative demand shock: $(1-\omega_t)u(c_t, n_t)$
  - ω_0 such that $\Delta Y_t = -1\%$ on impact
  - Reverts to $\omega = 0$ with persistence $\rho_\omega = 0.75$ at the quarterly level
- Unexpected, transitory, perfect foresight: a 'MIT' shock

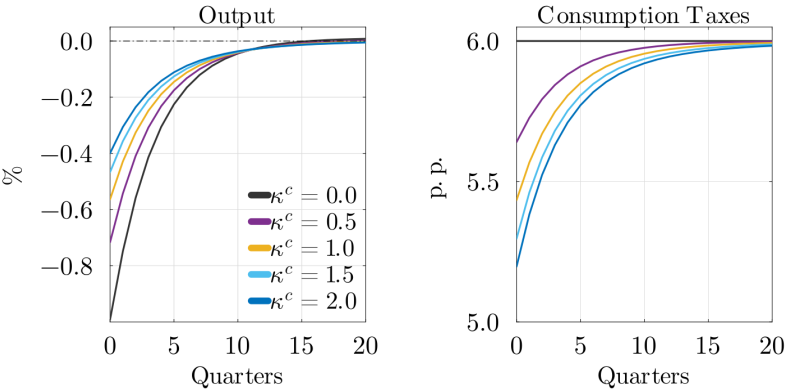
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  - Reverts to $\omega = 0$ with persistence $\rho_\omega = 0.75$ at the quarterly level
- Unexpected, transitory, perfect foresight: a 'MIT' shock
- Define a systematic rule: $\tau_t^c = \tau^c + \kappa^c \Delta Y_t$
  - κ^c captures the response of τ^c to the cycle
  - Vary $\kappa^c \in \{0, 0.5, 1, 1.5, 2\}$
  - Compare it to stabilization through G

Recession **No stabilization:** $\kappa^c = 0$

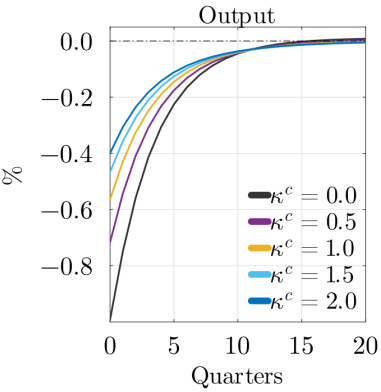


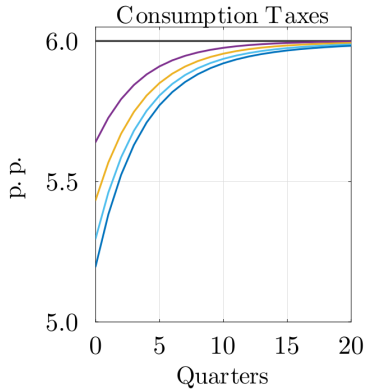
Recession **Stabilization** $\tau_t^c = \tau^c + \kappa^c \Delta Y_t$

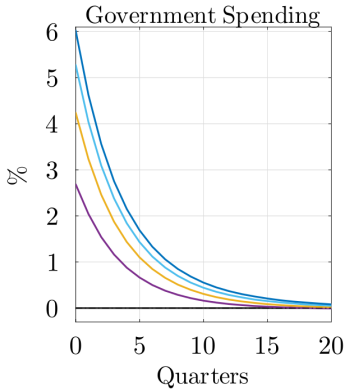


- Reducing the output fall by 50bp can be achieved with...
  - A consumption tax cut of 60bp

Recession **Stabilization** $\tau_t^c = \tau^c + \kappa^c \Delta Y_t$







- Reducing the output fall by 50bp can be achieved with...
  - A consumption tax cut of 60bp, or a spending shock of 5%

Welfare **Comparing τ^c to G**

- Let spending be valued by the household:

$$U(c_t, n_t, g_t) = \log c_t - Bv(n_t) + \chi \log g_t$$

Welfare **Comparing τ^c to G**

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- **Welfare**: Comparison between τ_c and G

$$\Delta_W^{\tau - g} = \sum_{t=0}^{\infty} \beta^t \left(\frac{G}{C_t^g} - \chi \right) \delta_t^g, \text{ with } \delta_t^g \text{ the size of the fiscal shock in terms of } G.$$

Welfare **Comparing τ^c to G**

- Let spending be valued by the household:

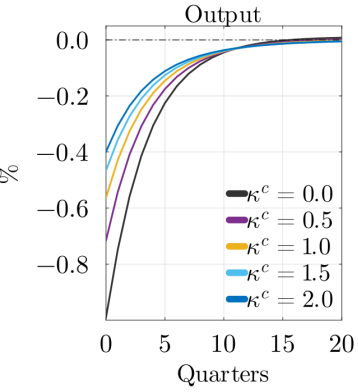
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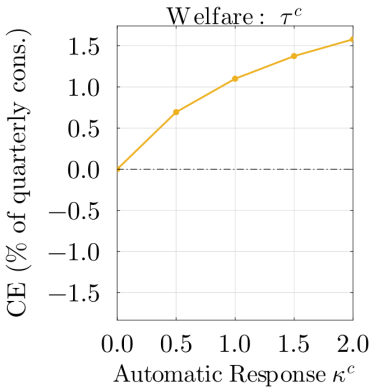
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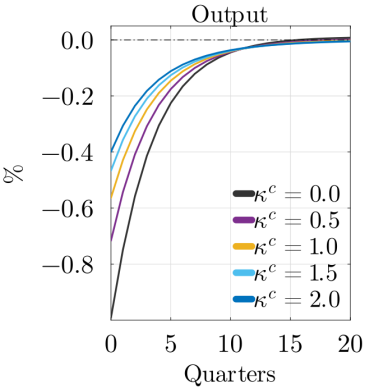
- What is χ? Two extreme cases
  1. Non-valued: $\chi = 0 \Rightarrow \Delta_W^{\tau-g} >> 0$
  2. Samuelson condition: Optimal G in steady state: $\chi = G/C$
     $\Rightarrow \Delta_W^{\tau-g} > 0$ in recessions, but small

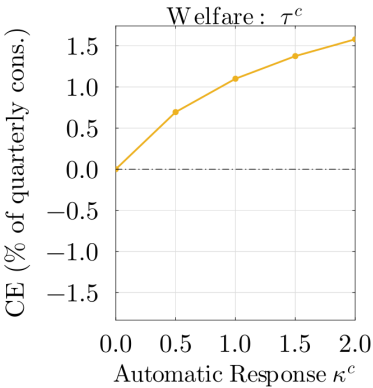
Recession Basic Welfare Computation

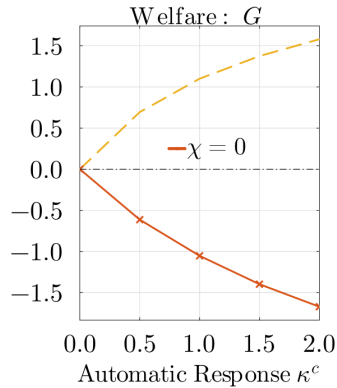




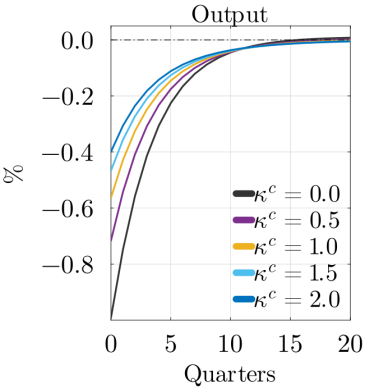
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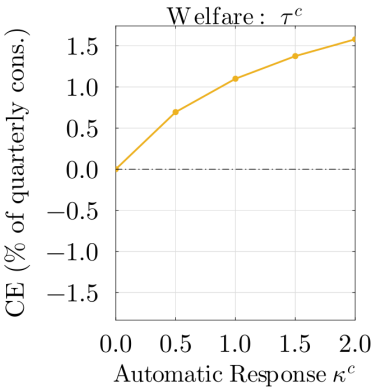


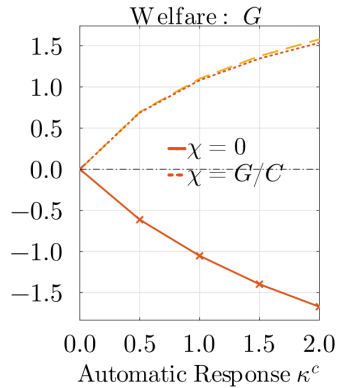




Recession Basic Welfare Computation







Welfare **Comparing τ^c to G**

- How valuable is extra public spending during recessions?

Welfare Comparing τ^c to G

- How valuable is extra public spending during recessions?

Global Insight Japanese war on deflation

Abe's stimulus brings back 'bridges to nowhere' spectre in Japan

While borrowing is easy, finding good ways to spend is not, economists say



Japan's 'maglev' railway will cut journey times between Tokyo and Osaka to just 67 minutes but will cost ¥9tn even if it opens on time and on budget in 2037 © Kyoda/Getty

Robin Harding in Tokyo

Published DEC 6 2019

In the late 1980s and 1990s, Japan spent ¥2.89tn to build 15 vast bridges between the islands of Honshu and Shikoku. The project was meant to stimulate a struggling economy — and it did — but tolls from the lightly used bridges fell short of the interest bill for construction debt. Though far from the worst excesses of that era, the Honshu-Shikoku project symbolises Japan's construction of "bridges to nowhere", leaving behind a legacy of concrete and public debt.

Conclusion

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- Consumption taxes as automatic stabilizers?
  - ⋆ Fiscal burden contingent on effectiveness of the stimulus
  - – Easy to implement, high pass-through, salient

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- Distributional gains?

Thank you!

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  - Replicate monetary policy by a combination of $\{\tau_t^c, \tau_t^n, \tau_t^k\}$ in a RANK
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- Wolf (2024)
  - Conditions under which changes in private spending and changes in public spending generate same general equilibrium effects
    - + Same path for aggregate demand
    - + Same path of future taxesr

Return

Extending the Equivalence

- Works with aggregate shocks and non-zero consumption tax
  - Let X_t denote a variable in t in absence of fiscal stimulus;
  - Fiscal costs defined as:

    $\hat{F}_t \equiv (\hat{G}_t - G_t) - \tau_t^c(\hat{C}_t - C_t)$ and $\bar{F}_t \equiv -(\bar{\tau}_t^c - \tau_t^c)\bar{C}_t - \tau_t^c(\bar{C}_t - C_t)$
    - The consumption tax cut that replicates the spending stimulus is given by:

$$\bar{\tau}_t^c = \tau_t^c \frac{\hat{Y}_t - \hat{G}_t}{\hat{Y}_t - G_t} - \frac{\hat{G}_t - G_t}{\hat{Y}_t - G_t}$$

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- **Equivalence** of any $\{\tilde{G}_t, \tilde{\tau}_t^c\}$ that deliver the same fiscal cost:

  $\tilde{F}_t \equiv (\tilde{G}_t - G_t) - (\tilde{\tau}_t^c - \tau_t^c)\tilde{C}_t - \tau_t^c(\tilde{C}_t - C_t)$
  - Finance spending shock with consumption taxes $\Rightarrow$ multiplier $= 0$
  - Full crowding-out of private consumption!

Return

Preferences

- Define $\mu \equiv u_c(c, n)$ and $\sigma^s = \frac{\partial \log \mu}{\partial \log c}$
- Define $R \equiv -\frac{u_n(c,n)}{u_c(c,n)}$ and $\sigma^n = \frac{\partial \log R}{\partial \log c}$

Return

Preferences

- King-Plosser-Rebello

$$U(c_t, n_t) = \frac{1}{1-\sigma}\left((c_t(1-n)^{\nu})^{1-\sigma} - 1\right)$$

delivers $\sigma^n = 1$ and $\sigma^s = \sigma$.

- Jaimovich-Rebelo

$$U(c_t, n_t; x_t) = \frac{1}{1-\sigma}\left(c_t - \frac{n_t^{1+\varphi}}{1+\varphi}x_t\right)^{1-\sigma}$$

where $x_t = c_t^{\gamma}$, delivers $\sigma_n = \gamma$ and $\sigma_s = \sigma\frac{\mu(1+\varphi)}{\mu(1+\varphi)-1}$.

- Adjusted CRRA

$$U(c_t, n_t; x_t) = \frac{c_t^{1-\sigma^s}}{1-\sigma^s} - x_t\frac{n_t^{1+\varphi}}{1+\varphi}$$

where $x_t = c_t^{\sigma^n - \sigma^s}$ is a reference point.

Return

Households **Working households**

- Individual state: asset a, discount factor β, productivity x, and employment $\eta \in \{\ell, u\}$

Return

Households **Working households**

- Individual state: asset a, discount factor β, productivity x, and employment $\eta \in \{\ell, u\}$
- Value function when employment "island" $\eta = \ell$

$$V_t(a, x, \ell, \beta; \mu_t) = \max_{c, a'} \left\{ \log c - B \frac{h^{1+\varphi}}{1+\varphi} + \beta \mathbb{E}_t \left[V_{t+1}(a', x', \eta', \beta'; \mu_{t+1}) | x, \beta, \ell \right] \right\} \quad \text{s.t.}$$

$$(1 + \tau_t^c) c + a' = a + y^\ell + y^k - \mathcal{T}_t(y^\ell, y^k) + T_t,$$

$$y^\ell = w_t x h, \quad y^k = r_t a, \quad a' \geq 0, \quad \mu_{t+1} = \Psi(\mu_t).$$

  - AR(1) process for discount factor, productivity and employment status
  - Flat capital and consumption taxes τ^k and τ_t^c, progressive loglinear labor tax (λ_t, τ^ℓ)
    Heathcote, Storesletten, and Violante (2017)

Return

Households **Unemployed households**

- Value function when in unemployment "island" $\eta = u$

$$V_t(a, x, u, \beta; \mu_t) = \max_{c, a'} \{\log c + \beta \mathbb{E}_t \left[V_{t+1}(a', x', \eta', \beta'; \mu_{t+1}) | x, \beta, u\right]\} \quad \text{s.t.}$$

$$(1 + \tau_t^c)c + a' = a + y^k + \mathcal{B}_t(w_t x h) - \mathcal{T}_t(0, y^k) + T_t,$$

$$y^k = r_t a, \quad a' \geq 0.$$

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$$(1 + \tau_t^c)c + a' = a + y^k + \mathcal{B}_t(w_t x h) - \mathcal{T}_t(0, y^k) + T_t,$$

$$y^k = r_t a, \quad a' \geq 0.$$

  - Unemployment benefits function of hourly wage

    Kekre (2022)

$$\mathcal{B}_t(w_t x h) = \zeta \min\left(\mathcal{R} w_t x h, \overline{ui}\right) + \chi w_t x h$$

    + ζ to match fraction of recipients, $\mathcal{R}$ the replacement rate, $\overline{ui}$ the UI cap
    + χ to capture household labor income received while in unemployment

  - AR(1) process for discount factors, productivity and employment status

Return

Firms and Labor Market

- Standard two-layer structure
  Hagedorn, Manovskii, and Mitman (2025)

Return

Firms and Labor Market

- Standard two-layer structure
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- Representative final good producer
- Intermediate good producers indexed by k
  - Linear technology: $y_{kt} = n_{kt}$, flexible prices $\rightarrow$ constant wage w

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Firms and Labor Market

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  Hagedorn, Manovskii, and Mitman (2025)
- Representative final good producer
- Intermediate good producers indexed by k
  - Linear technology: $y_{kt} = n_{kt}$, flexible prices $\rightarrow$ constant wage w
- Unions indexed by j
  - Demand same labor from all working households and aggregate it linearly
  - Nominal rigidities a la Rotemberg

Return

Unions **Phillips Curve**

- Continuum of unions j demanding labor h_{ji} from households i
  - Labor demand: $n_{jt} = \left(\frac{W_{jt}}{W_t}\right)^{-\epsilon} Y_t$
  - Nominal wage W_{jt} s. to utility cost: $\frac{\Theta}{2}\left(\frac{W_{jt}}{W_{jt-1}} - \bar{\Pi}\right)^2 Y_t$
  - Atomistic union: take as given total consumption and total hours of working households

$$J_t(W_{tj-1}) = \max_{W_{jt}} \left\{ \int \left[\frac{W_{jt}}{P_t} \frac{(1 - MTR_i)x_i}{1 + \tau_{ct}} - \frac{v'(h_i)}{u'(c_i)} \right] h_j di - \frac{\Theta}{2}\left(\frac{W_{jt}}{W_{jt-1}} - \bar{\Pi}\right)^2 Y_t + \beta J_{t+1}(W_{tj}) \right\}$$

Return

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- **Phillips curve** collapses to Erceg et al (2000) with no heterogeneity

$$\left(\Pi_t^w - \bar{\Pi}\right)\Pi_t^w + \frac{\epsilon - 1}{\Theta}\frac{w}{\bar{x}_t}\int \frac{(1 - MTR_i)x_i}{1 + \tau_{ct}} di = \frac{\epsilon}{\Theta}\frac{1}{\bar{x}_t}\int \frac{v'(h)}{u'(c_i)} di + \beta\left(\Pi_{t+1}^w - \bar{\Pi}\right)\Pi_{t+1}^w \frac{Y_{t+1}}{Y_t}$$

Return

Government

- Monetary authority follows a Taylor rule: $1 + i_t = (1 + \bar{i}) \left(\frac{\Pi_t}{\bar{\Pi}}\right)^{\phi_\Pi}$
- Fiscal authority faces a standard borrowing constraint

$$G_t + (1 + r_t)D_t + T_t + \int \mathcal{B}_t(w_t x)d\mu_t = D_{t+1} + \int \mathcal{T}_t(y_t^\ell, y_t^k)d\mu_t$$

- Fiscal rule with parameter Φ_D for public debt, λ_t clears the budget constraint
  Uhlig (2010)

Return Fiscal rule

Government

- Monetary authority follows a Taylor rule: $1 + i_t = (1 + \bar{i}) \left(\frac{\Pi_t}{\bar{\Pi}}\right)^{\phi_\Pi}$
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- Fiscal rule with parameter Φ_D for public debt, λ_t clears the budget constraint
  Uhlig (2010)
  - $\Phi_D = 0$ for constant debt, all adjustment in tax level
  - $\Phi_D \to 1$ for constant taxes, all adjustment in debt

    $D_{t+1} = (1 - \phi_D)D + \phi_D \left(\hat{G}_t - \tau^k r_t A_t - \tau_t^c C_t - \mathcal{R}_t^\ell\right)$, where
    - $\hat{G}_t$ denotes total spending, including debt repayments
    - $\mathcal{R}_t^\ell$ denotes fiscal revenues at steady-state labor tax schedule

Return Fiscal rule

Steady State **Households**

- Quarterly model calibrated to liquid wealth

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- Stochastic $\beta \in \{\bar{\beta} - \Delta,\ \bar{\beta},\ \bar{\beta} + \Delta\}$, duration of 50 years

  Krusell and Smith (1998)
  - $\bar{\beta}$ s.t. $r \equiv 3.5\%$ annually
  - Δ s.t. top-quintile liquid wealth $\approx 80\%$ (SCF)

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- Labor supply
  - B to match hours $h = 1/3$, Frisch $\varphi^{-1} = 0.4$
- Productivity $(\rho_x, \sigma_x) = (0.989, 0.287)$

  Chang and Kim (2007)

Return Fiscal rule

Steady State **Firm and government**

- Technology: $\varepsilon = 7$, $\Theta = 200 \rightsquigarrow$ Phillips curve slope $\varepsilon/\Theta = 0.035$
  Galí and Gertler (1999)

- Government
  - Capital tax $\tau^k = 35\%$, consumption tax $\tau^c = 6\%$
    Labor income tax progressivity $\tau^\ell = 0.06$, transfers to match estimates from CPS
    Chen, Imrohoroglu, and Imrohoroglu (2007), Heathcote, Storesletten, and Violante (2017), Ferriere, Grübener, Navarro, and Vardishvili (2023)
  - Fiscal revenue-over-output 22%, debt $D/Y = 100\%$, residual spending $G/Y = 11\%$

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    $\chi = 0.24$ to match $C_u/C_e \approx 70\%$
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  - Automatic responses of inflation and debt: $\Phi_\Pi = 1.5$, $\Phi_D = 0.75$

Return Fiscal rule

Unemployment Steady State and Business Cycles

- Job finding rates and separation rates across hourly wage distribution

  Mueller (2017)

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- Steady State
  - Job finding rates constant, separation rates falling in hourly wage/productivity x
  - Average unemployment rate at 4.2%

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- Okun's law: Okun coefficient $c_{OK} = 0.5$
  Ball, Leigh, and Loungani (2017)
  - Job finding rates decrease (a lot) equally across households
  - Job separation rates decrease (a bit), higher elasticity for high-x households
    + Functional forms: additive fall in separation rates in recession delivers the pattern

Fiscal Rule

- Public debt adjusts as a function of Φ_D

$D_{t+1} = (1-\phi_D)D + \phi_D \left(\hat{G}_t - \tau^k r_t A_t - \tau_t^c C_t - \mathcal{R}_t^\ell\right)$, where

  - $\hat{G}_t$ captures total government expenditures, including debt repayments

    $\hat{G}_t = G_t + T_t + \mathcal{U}_t + (1+r_t)D_t$

  - $\mathcal{R}_t^\ell$ captures fiscal revenues at steady-state labor tax schedule

    $\mathcal{R}_t^\ell = w_t L_t - \lambda \int (w_t x h_t(a,x,\eta,\beta))^{1-\tau^\ell} d\mu_t(a,x,\eta,\beta)$

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Steady State **Unemployment**

- Job finding rates and separation rates across hourly wage distribution

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- Job finding rates are constant in the distribution
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  - Monthly finding rate of $0.32 \Rightarrow \pi_\eta(\ell|u) = 0.691$

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- Separation rates are falling in hourly wage/productivity x
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  - Monthly separation rates of $\approx 1.4\%$ and 0.7% below and above median, respectively

  $\Rightarrow$ $\pi_\eta(u|\ell, x) = \phi_0 x^{\phi_1}$, with $\phi_0 = 0.029$ and $\phi_1 = -0.446$

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- Average unemployment rate at 4.3% with unequal incidence in the distribution

Return

Unemployment and the Business Cycle

- Okun's law type of relation between output and unemployment
  - Okun coefficient $c^{OK} = 0.5$

    Ball, Leigh, and Loungani (2017)

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  - Elasticity of job finding rates to aggregate unemployment of -0.6
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- Job separation rates decrease with ΔY_t
  - *Elasticity* of separation rates to aggregate unemployment larger for above-median workers
    Mueller (2017)
    + Homogeneous additive increase in separation rates

Unemployment and the Business Cycle **Okun's law**

- Finding and separation rates distribution depend on U_t Mueller (2017)
  - Finding rate responds homogeneously (level, independent of x) to ΔY_t

$$\pi_{\eta,t}(\ell \,|\, u) = \pi_{\eta}(\ell \,|\, u) + \bar{\phi}_e \, \Delta Y_t$$

  - Separation rate elasticity increases with ΔY_t

$$\pi_{\eta,t}(u|\ell, x) = \pi_{\eta}(u|\ell, x) - \bar{\phi}_u \Delta Y_t x^{-\phi_{u,x}}$$

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  - Joint calibration:
    + $\bar{\phi}_e$ s.t. finding elasticity to $U \approx -0.6$
    + $\phi_{u,x} = 0$ elasticity of separation rates larger for above-median workers
    + $\bar{\phi}_u = 0.32$ to get $c_{OK} = 0.5$

Return